

# **MegaMatcher Accelerator 12.4**

## **Manual**

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# Overview

## Introduction

MegaMatcher Accelerator is a solution for server-side part of a large-scale biometric systems, such as ABIS (Automated Biometric Identification System), duplicate search, etc. It includes very fast matching algorithms for fingerprints, faces and irises – up to 1,2 billion (1,200,000,000) matches per second depending on edition. The solution can also scale to more than one machine physical limitations to form a cluster suitable to meet any performance and capacity requirements.

For more information about the product please visit: <http://neurotechnology.com/megamatcher-accelerator.html>.

## What's new

### 12.4 Release

Updated algorithm versions.

Changed fingers to use large template by default (better accuracy, but requires more RAM).

Fixed iris capacity counting for very large systems with Irises Extended license.

### 12.3 Release

Added new gRPC-based API which removes requirements to use Neurotechnology SDK for communication. Client libraries may be generated and used in practically any programming language.

Fixed issues where replicated partitions were kept in memory longer than needed.

Improved performance when replicated node leaves or joins cluster.

### 12.2 Release

Allow to take full advantage of quality improvements in iris templates extracted using NeurotecSDK 12.2.

Extended and improved template replication in memory functionality useful for high availability. See Replication topic for more details about this feature.

Changed default configuration value to convert fingerprint templates to small representation when loading to memory. This results in significantly less RAM usage while not having noticeable impact on accuracy in most cases.

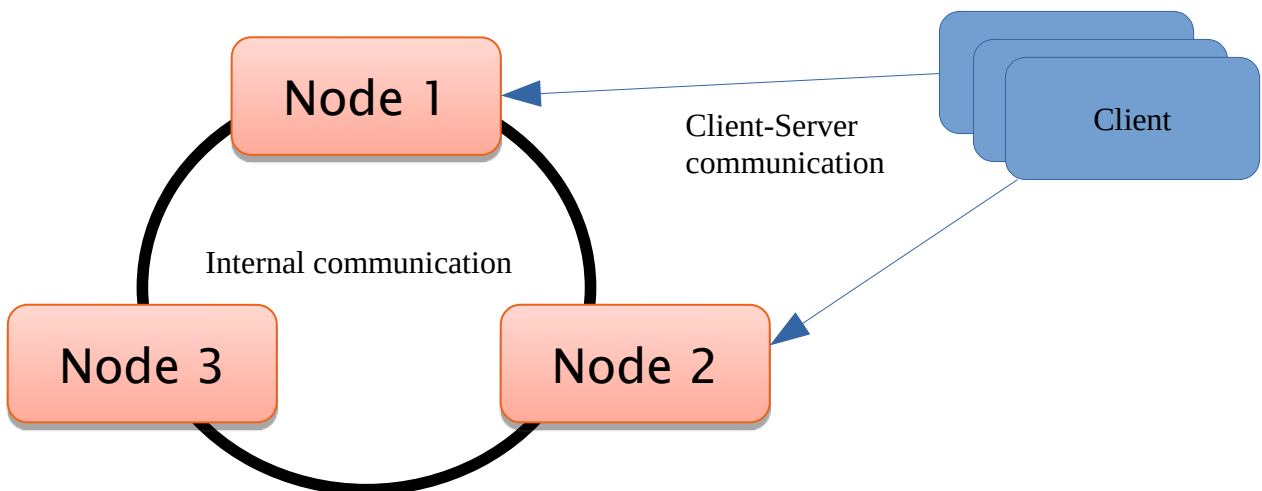
## 12.0 Release

The latest MegaMatcher Accelerator 12.0 includes major upgrades to fingerprint, face, and iris algorithms for fast and highly accurate identification using single or multiple biometric modalities.

MegaMatcher Accelerator Extreme edition can now match up to 1.2 billion fingerprint, face or irises per second without a GPU on a single server. This greatly simplifies deployment without requiring installation of special hardware. And, most importantly, this high speed does not come at a cost of reduced accuracy - the Extreme edition's recognition accuracy is the same as for other slower MegaMatcher Accelerator editions.

## Cluster architecture

MegaMatcher Accelerator (MMA) is composed of one or more worker nodes which automatically distribute biometric database (gallery) and requests from clients between themselves (see Illustration 1: MMA cluster architecture).



*Illustration 1: MMA cluster architecture*

There are no special or master nodes in the system, all of them are equal and each of them can carry out the same tasks. Each node has a consistent copy of the database stored on disk, so there is no requirement for specialized storage solutions such as SAN. This design eliminates single point of failure and simplifies operation and maintenance.

Clients connect to the cluster to perform operations such as enroll or identify. Any single node connection is enough to perform any task, but more can be specified for automatic failover.

MMA is a strongly consistent system. In terms of CAP theorem, it is a CP. Meaning, that in case of network partition the system prefers data consistency and correctness instead of allowing to make possibly invalid changes.

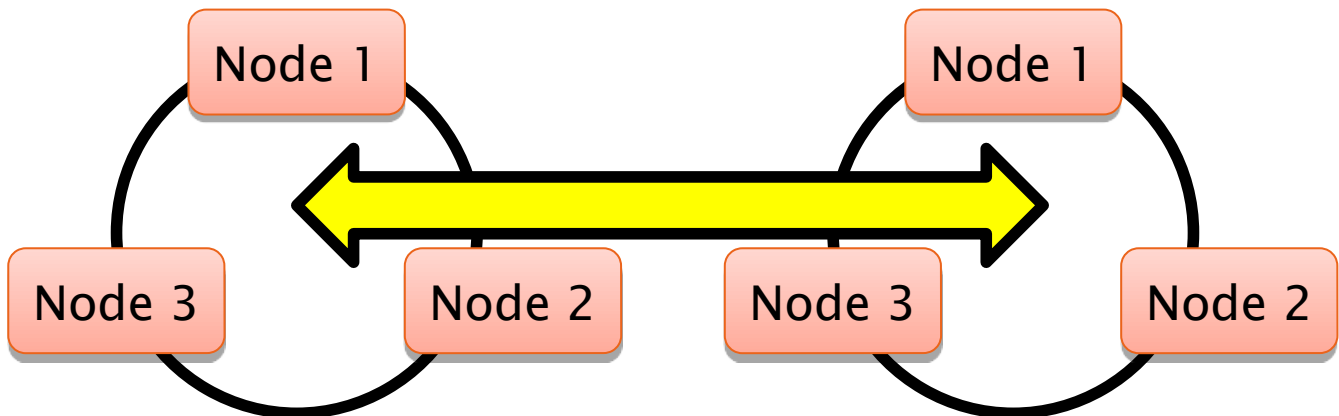
## Federation (multiple clusters connected)

MegaMatcher Accelerator cluster requires that nodes are connected closely usually in LAN (Local Area Network). Some projects are required to setup a mirror system in a remote site which can be in distant location, such as another city. Connectivity between such remote areas may not be always available.

MMA Federation allows to connect multiple clusters so that changes from one are propagated to the other. If change propagation is configured both from cluster A to B and vice versa, both clusters eventually will have the same data.

Federation, different from cluster internal communication, uses asynchronous change propagation. It means that when change happens in cluster A, cluster B may not see the change immediately. There may be some delay, which is usually very short. It is also possible that the clusters are disconnected for some time, and then when connection is available again they resynchronize from the point where they last stopped.

As the changes are not immediately synchronized, it is possible that some conflicts may occur. For example, the same person is enrolled in different cities in very short time interval or during the period when there is no connectivity. Federation system is designed to detect and give administrator opportunity to fix such conflicts should they occur. Such conflicts will not occur in active-passive systems where only one cluster is allowed to make changes.

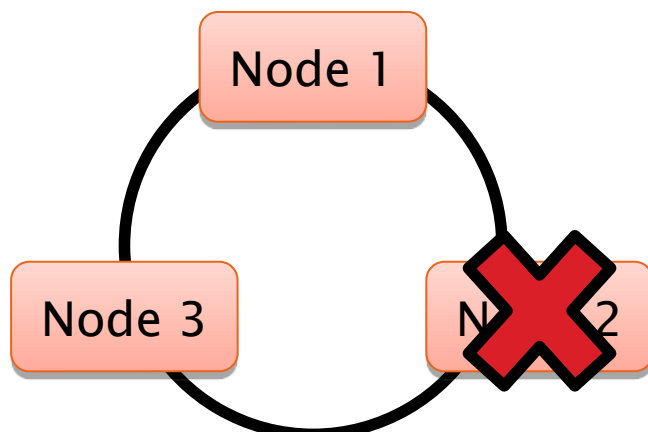


*Illustration 2: Federation connection between clusters*

## Failover

Failover functionality allows to restore cluster operation after event of one or more nodes abnormally leaving the cluster. Node failure may be caused by many reasons: node's hardware failure, network failure, software issue, etc. In any case one or more nodes become unavailable.

MMA cluster automatically detects such events and takes action necessary to restore normal



*Illustration 3: Failure scenario: one of cluster nodes becomes unavailable*

cluster operation. Each node loads just the minimum amount of data from disk which is required to replace the missing nodes. All this happens automatically without requiring administrator actions.

During the failover data loading process the cluster cannot process identify requests which require full database to be loaded. They are queued and processed after the failover procedure finishes.

It should be noted that it is not reasonable to perform failover if too many nodes become unavailable. If the remaining nodes cannot fit the whole database in memory, the cluster will not function properly and is likely to become unresponsive or even crash. Therefore a minimal number of nodes for cluster to function properly should be defined in the configuration. If there are fewer nodes than required, the remaining nodes would simply wait for more nodes instead.

## **Scalability**

MMA cluster is scalable both horizontally (adding additional nodes to cluster) and vertically (using more powerful hardware) depending on project needs. For smaller systems, vertical scalability may be easier to setup (especially when there is only one node). For larger systems, horizontal scalability is usually preferred.

If you need to adapt MMA for increased performance or capacity requirements, please don't hesitate to contact [support@neurotechnology.com](mailto:support@neurotechnology.com) for consultation on most optimal options.

### **Horizontal scalability**

MMA can be scaled horizontally by installing and adding new nodes to the cluster. This action can be performed without downtime and even without interrupting normal system operation. Please note that newly added node should have similar hardware performance characteristics as the existing ones.

### **Vertical scalability**

Vertical scalability is possible in case one or both of the following limitations is increased to meet the new requirements:

- Hardware physical limitations. In case processing speed is a bottleneck, usually hardware with better CPU is required. More RAM may be needed if there is a need for more capacity.
- License limitations. There are several MegaMatcher Accelerator editions available which provide different capacity and performance limitations. Please refer to Neurotechnology website for up-to-date details: <http://neurotechnology.com/megamatcher-accelerator.html> . If hardware does not need to be upgraded, license limitations can be increased without reinstalling the system.

## Read-only Nodes

If having full database copy for each MegaMatcher Accelerator node is not desirable – Read-only nodes can be added to the cluster. These read-only nodes will read data from other nodes databases and will not write to any database.

In order to configure node to be read-only add `--no-database` during installation step:

```
# ./neurotec-biometric-node-ctl.sh --no-database install
```

More complex database configurations are also supported (e.g. external cluster of databases). Please consult with [support@neurotechnology.com](mailto:support@neurotechnology.com) for configuration instructions.

## Replication

MMA allows to configure replication to load more than one copy of data in memory (on different machines). When dealing with some MMA nodes failing in cluster (because of machine breaking down, network unavailable, software issue, etc.), the remaining nodes can load parts of database previously handled by failed nodes and continue work. When there is one copy of template data in memory, missing parts need to be loaded from disk which takes significant time. With template replication  $> 1$ , there is another copy of templates already loaded in memory. Therefore, remaining nodes do not need to access disk and only perform very fast in memory data switch.

Before enabling replication, it is important to do capacity planning. If two copies of templates are loaded (Replication = 2), it means that twice as much RAM is required, and also twice as much of capacity limits specific for MMA license are used. Capacity limits should not be exceeded even when failure happens and remaining nodes load the missing parts. If too many of nodes are not available for the system to work without exceeding limits, cluster should be configured to stop and wait for more nodes instead. Configuration parameter `NodesToStop` should be used for this.

For example, if you have 6,000,000 fingerprints, that means you need 2 units of MegaMatcher Accelerator Standard – minimum number of nodes to fit all data to memory. To set Replication to 2 you need to have at least 3 nodes. `NodeToStop` could be 1 in this case as such number is not enough to fit all data to memory.

Parameter for Replication can be set using Web Configuration tool. The parameter can be changed dynamically on running cluster without a need to stop/restart.



## Installation

MegaMatcher Accelerator depending on chosen edition and configuration may come pre-installed. In such case there is no need to perform any steps described in this section.

MegaMatcher Accelerator is distributed as Docker containers which is suitable for a lot of different linux environments and is often familiar for experienced administrators. Yet the distribution package provides scripts which automate installation steps and experience with Docker is not required.

## Hardware setup

Make sure your machine meets hardware requirements as described in Neurotechnology website: <http://neurotechnology.com/megamatcher-accelerator-system-requirements.html>. Other hardware configurations may also be possible, please consult with [support@neurotechnology.com](mailto:support@neurotechnology.com).

When using Virtual Machine appliances, make sure that CPU features such as SSE 4.2 are allowed on guest machine.

Configure RAID, if applicable.

## Install base operating system

A linux based operation system supporting recent versions of Docker is required. Below are install instructions for recommended OS distributions but alternatives could be chosen as well.

1. Install linux distribution. We recommend Debian 10.2.
2. Install Docker. For Debian follow instructions in <https://docs.docker.com/install/linux/docker-ce/debian/>.
3. Install Docker Compose. Follow instructions in <https://docs.docker.com/compose/install>.

## Run licensing service

All units of the MegaMatcher Accelerator require a license. Licensing server needs to be launched on every machine where the dongles holding the required licenses are attached.

Steps required to setup start *activation* server:

1. Download `Activation_12_1_Linux_x86_64_<release_date>.zip` archive and extract on target machine. Note: If licenses are used from USB dongles, Activation should be deployed directly on host machine, not virtual machine or container.
2. Activation server is controlled by npgadmin service which allows easy configuration through Web Control Panel.

1. For Debian based systems, helper script may be used to register npgadmin for automatic startup using systemctl. Use `# startup_npgadmin.sh register` command to register npgadmin in systemctl service list. After that, the service can be controlled through systemctl tool.
2. For other systems it is needed to create a script which launches npgadmin executable.
3. Navigate to Web Control Panel > Licensing page to choose the licensing mode and view current status.

## MMA Installation Wizard

MegaMatcher Accelerator installation package contains Docker images as well as helper scripts which are required for installing and also performing other administration tasks.

1. Download MMA installation package from Neurotechnology website. If Internet connection on installation system is not available, base Docker images can be downloaded from the same page as well for completely offline installation.
2. Unzip the downloaded files. If base Docker images are used, make sure they are extracted to the same directory.
3. Change directory to MegaMatcher\_Accelerator\_12\_1\_Cluster. MMA installation wizard can be started with help instruction to list all available commands using:  
`# ./neurotec-biometric-node-ctl.sh -h`

Note: if MMA is installed to be integrated with MegaMatcher ABIS product, an additional flag -a needs to be specified. Please refer to MegaMatcher ABIS documentation for more details.

## Install MMA Docker images

Invoke MMA Installation Wizard without arguments and follow instructions on screen:

```
# ./neurotec-biometric-node-ctl.sh -h
```

## Uninstall

Purge command can be useful if there is a need to completely remove MegaMatcher Accelerator Cluster components and all data created by the system.

It is also needed in case of partial or uncompleted installations which have to be cleaned up, so that the new installation could start from clean state.

The following command should be used to purge all components and data:

Invoke MMA Installation Wizard with purge command:

```
# ./neurotec-biometric-node-ctl.sh purge
```

## Finalizing installation

After installation the node is still not running and not accessible. Please refer to Operating chapter for details how node should be started and other day to day operation work be performed.

## Migrating from previous major version of MMA

It is not possible to directly upgrade from previous major version of MMA, multiple steps need to be performed.

1. Create a backup of previous MMA installation database. It is recommended to check if backup is correct by first restoring it to some testing environment.
2. Install the MMA as described in Replication part of this document.
3. Start node
4. In Web Configuration go to Database section and in Restore database section choose previously saved backup file and start restore procedure.

Please note: if you have very old biometric templates and still have original images used for extraction, instead of restoring old templates it is recommended to reenroll the images using the newest version of template extractor. This allows to achieve the best recognition accuracy.

## Web Configuration

MegaMatcher Accelerator includes administration tool which can be accessed using web browser. Note that the MMA node has to be installed (see Replication) and started (see Start node) before using the described functionality. Navigate to [http://ip\\_address](http://ip_address) (replace ip\_address placeholder with the actual IP of installed server) and enter the default credentials: Admin / Admin.

## Dashboard

The dashboard provides overview of the whole cluster status. It includes status information for each node – whether it is loading data, is failed, or running properly. Also important statistics, such as how much of database capacity is used is shown. Administrator can take corrective actions if needed, such as adding additional nodes to increase capacity or reinstall broken node on a different machine.

## Administrative tasks page

All non-biometric tasks performed in MMA, such as starting a cluster or changing configuration, are considered administrative and logged in the system. This is useful both to find an outcome of long-taking command (any errors are also displayed here), or to examine what changes have happened in the system.

### Administrative tasks

Home / Administrative tasks

**Filter**

Created

from

to

Ended

from

to

Status

--

Phrase

Filter

Reset

**Administrative tasks list**

| # | Title         | Created at              | Progress    | Status   | Ended at                | Error message |
|---|---------------|-------------------------|-------------|----------|-------------------------|---------------|
| 6 | Start cluster | 2017-06-22 10:22:05.739 | <div></div> | Finished | 2017-06-22 10:22:05.771 |               |
| 5 | Start cluster | 2017-06-22 10:02:37.155 | <div></div> | Finished | 2017-06-22 10:02:37.190 |               |

Illustration 4: Administrative tasks page

## Node page

Node page provides detailed information about the node connected to. Node log is shown which can be downloaded as short (tail of log), or full.

The Node page displays the status of a specific node. The 'Node' section shows the status as **RUNNING** and the ID as **88786075971**. Below this, the 'Log' section provides a list of messages. There are buttons for 'Download short log' and 'Download full log', and a checkbox for 'Sticky scrollbar'.

| Timestamp           | Message                                        |
|---------------------|------------------------------------------------|
| 2017-06-22 10:22:53 | Info: Partition 4088 loaded into Engine to 102 |
| 2017-06-22 10:22:53 | Info: Partition 4088 loaded into Engine to 153 |
| 2017-06-22 10:22:53 | Info: Partition 4088 loaded into Engine to 204 |
| 2017-06-22 10:22:53 | Info: Partition 4088 loaded into Engine to 205 |
| 2017-06-22 10:22:53 | Info: Partition 4091 loaded into Engine to 51  |
| 2017-06-22 10:22:53 | Info: Partition 4091 loaded into Engine to 102 |

Illustration 5: Node page

## Cluster page

Cluster page shows a list of connected nodes, including their addresses and statuses.

The Cluster page displays the status of the cluster as **READY**. Below this, the 'Nodes' section provides a list of connected nodes with their IDs, cluster IDs, IP addresses, and stages.

| Node ID     | Cluster ID                           | IP address  | Stage          |
|-------------|--------------------------------------|-------------|----------------|
| 71606206788 | 705290E4-38F6-A043-8D58-2A25AAE6007F | 172.16.1.68 | <b>RUNNING</b> |
| 88786075971 | 705290E4-38F6-A043-8D58-2A25AAE6007F | 172.16.1.67 | <b>RUNNING</b> |

Illustration 6: Cluster page

## Metrics page

System information about network, CPU, memory, disk usage is shown in *Metrics window*.

If you need more detailed information, press *More metrics*. This will open *Graphite Browser*, the tool providing detailed system parameters information.



*Illustration 7: Metrics page*

## Developing solution

MegaMatcher Accelerator is a server-side solution which executes tasks requested by client application or other systems (referred to as Client later in this document). Client can be developer using one of Neurotechnology Biometric SDKs: MegaMatcher, VeriFinger, etc. For more information including download links please visit <http://neurotechnology.com/megamatcher.html>.

Client can perform many steps needed to generate high quality biometric template: acquire biometric samples from scanners or images, perform quality check, segment, extract the actual biometric template. SDKs documentation and included samples and tutorials instruct how to perform the described scenarios.

MegaMatcher Accelerator support is tightly integrated into Neurotechnology Biometric SDKs. It is easy to setup a connection using NBiometricClient, and send tasks to MMA in the same way as local biometric engine is used.

## Connection to MMA

Setting up connection to MMA only requires to create a NClusterBiometricConnection which describes host name of MMA. It is possible to specify connection port (AdminPort property) if not a default one is used on server.

```
var connection = new NClusterBiometricConnection
{
    Host = server,
    AdminPort = port
};
biometricClient.RemoteConnections.Add(connection);
```

In NClusterBiometricConnection more than one address of MMA node to connect to may be specified. This will enable automatic failover in case it is impossible to communicate with the node.

## Tutorials

Neurotechnology SDK includes tutorials which demonstrate small source code fragments needed to perform common scenarios. Tutorials are available in NeurotecSDK/Tutorials/Biometrics directory. There is a subdirectory for each supported programming language. All programming languages contain the same tutorials (only naming convention and other minor details are adapted to language but not the actual functionality). Table below introduces tutorials closely related to MMA usage scenarios.

| Tutorial       | Description                                                             |
|----------------|-------------------------------------------------------------------------|
| EnrollToServer | Demonstrates how to enroll a biometric templates stored in file to MMA. |

| <b>Tutorial</b>  | <b>Description</b>                                                        |
|------------------|---------------------------------------------------------------------------|
| IdentifyOnServer | Demonstrates how to identify a biometric templates stored in file to MMA. |

These tutorials only demonstrate the simple case when using already created template. It is also possible to create a template from images, devices, etc. NBiometricClient handles this automatically: if template has to be extracted from image, this is performed by the NBiometricClient on client side implicitly and the created template is sent to MMA. Please refer to other tutorials and the documentation for details.

## Samples

A typical scenario for client application is to collect subject's biometric data from scanner and performing the necessary biometric data quality checks, extract a biometric template and enroll it to a remote server (MegaMatcher Accelerator).

Neurotechnology Biometric SDK includes a sample application Abis Sample which demonstrates a typical MMA client application. Source code for this application is provided so the user can adapt it to her needs or to check how to use the certain functionality of the SDK. Abis Sample is provided for C++ (AbisSampleWX.exe), C# (AbisSampleCS.exe), VB.NET (AbisSampleVB.exe) and Java (abis-sample.jar) programming languages. Windows binaries are saved in Bin\Win32\_x86 and Bin\Win64\_x64, Java - Bin\Java, Linux - Bin\Linux\_x86 and Bin\Linux\_x86\_64 folders of the Biometrics SDK. Source code for this sample application is saved in these locations:

- Samples\Biometrics\CPP\AbisSampleWX (for C++)
- Samples\Biometrics\CS\AbisSampleCS (for C#)
- Samples\Biometrics\Java\abis-sample (for Java)
- Samples\Biometrics\VB\AbisSampleVB (for VB.NET)

Before starting using Abis Sample, it is recommended to read the user guide which is provided in Biometrics SDK documentation (see section Samples → Biometrics → ABIS).

Abis sample is a big application covering many areas and functionality of Biometrics SDK and providing communication with a remote server. If you are just starting with MegaMatcher Accelerator, it is a good idea to start with tutorials. These simple command line programs demonstrate the main functionality such as templates enrollment, communication with the remote server or subjects identification using simple code.

## Testing performance

When measuring performance of the system in development stage, the most accurate results can be calculated when large database of real production biometric data is used. However, in many cases such



data is not available and database for testing has to be created artificially. We provide several best practices which help to achieve more accurate speed testing results:

- Send identification tasks concurrently. If the tasks are sent one-by-one, the server has nothing to do when results are received by the client and new tasks request is sent. Therefore it is necessary to maintain multiple tasks executing in parallel, always keeping the server fully loaded.
- Use large enough database: at least 20% of the server capacity is recommended. If database of such size is not available, it is possible to repeat the gallery several times.
- Make sure templates sent for identification do not match with large part of templates in the database (gallery). In real scenarios, the database contains lots of templates from different subjects, and only very few (usually just one) of the same subject. Same subject's templates are matched much more slowly.
- When using fingerprints, speed is affected by the size of fingerprint image and number of minutiae the fingerprint contains. It is recommended to test with images which are similar to the ones used in the production system.

# Operating

## Start node

Invoke MMA Installation Wizard with start command:

```
# ./neurotec-biometric-node-ctl.sh start
```

## Stop node

Invoke MMA Installation Wizard with stop command:

```
# ./neurotec-biometric-node-ctl.sh stop
```

## Updating node configuration

First invoke MMA Installation Wizard with config command:

```
# ./neurotec-biometric-node-ctl.sh -l <address> -p <port> config
```

If *-l* or *-p* flag is defined, config command will also update these parameters, else only node and additional nodes IP update wizard will start.

Next invoke MMA Installation Wizard with update command, which will update node configuration files and update docker containers:

```
# ./neurotec-biometric-node-ctl.sh update
```

## Measure disk read/write speed

Invoke MMA Installation Wizard with io-test command, will measure disk write and read speeds for 1GB size file:

```
# ./neurotec-biometric-node-ctl.sh io-test
```

## Measure network speed between nodes hosts

First invoke MMA Installation Wizard with net-speed command with arguments *--net-test-ip* <define other node host IP address> and *--net-test-port* <define port to use for net-test>. Executing command will ask accessed host password and then measures bandwidth speed between nodes hosts:

```
# ./neurotec-biometric-node-ctl.sh io-test --net-test-ip <address>
--net-test-port <port>
```

## Configure parameters

MMA parameters can be configured in Accelerator Page in Web Configuration.

## **Nodes to start**

Only after a number of nodes specified in Nodes to start configuration parameter join cluster, cluster switches to running state to load biometric database and allow processing of requests.

## **Nodes to stop**

When some nodes crash or are removed and total amount of left nodes reaches count specified in Nodes to stop, cluster is stopped. Recommended value for Nodes to stop field is more than half of running nodes in a cluster (at least  $50\% + 1$  of running nodes). When deciding on minimal number of nodes, it is necessary to calculate that remaining nodes should not exceed limitations of licensing (such as database capacity) and hardware (such as available RAM).

## **Max task count**

Represents maximum number of running tasks. Server will not accept new tasks if number of currently running tasks reached maximum task count until some tasks are completed.

## **Accelerated matching mode**

Defines which modalities should be matched in accelerated mode. The choice should depend on available licenses. Please note that other modalities are also matched, but accelerated modalities are primary and at least one instance of accelerated modalities requires to be present in each template in order to ensure expected performance.

## **Accelerated fingers matching maximal rotation**

Sets fingerprints rotation angle in degrees. If fingerprints will be rotated more than a specified angle, it will not be matched by MegaMatcher Accelerator.

This field can contain values in range  $[0; 180]$  (degrees). Default: 180 (full rotation).

## **Accelerated fingers matching speed level**

Sets Accelerator's matching speed for accelerated fingerprints matching. If user requires faster matching with decreased quality then Highest fingerprints matching speed (8) should be used. If user prefers better matching accuracy (better ROC) then Slowest matching speed (1) should be used. Value between the two represent a trade-off. Default value: 1.

## **Use fingers adaptive speed**

This field sets automatically optimal fingerprint matching parameters in the cases when the DB contains templates with mixed number of fingerprints and/or mixed normal template with old template formats (i.e. with and without quality) and/or ISO templates. When Off, matching speed will be the same as set with Accelerated fingers matching speed level parameter. When On, Accelerator will adapt

matching speed depending on records contained in the template. It is recommended to use On only when user is sure that number of records in templates is different.

## Accelerated irises matching maximal rotation

Sets the maximum iris rotation angle in degrees. If irises will be rotated more than a specified angle, it will not be matched.

This field can contain values in range [0; 180] (degrees).

Default: 15.

Note: Irises matching speed depends on rotation angle: the higher the rotation angle, the slower the matching speed.

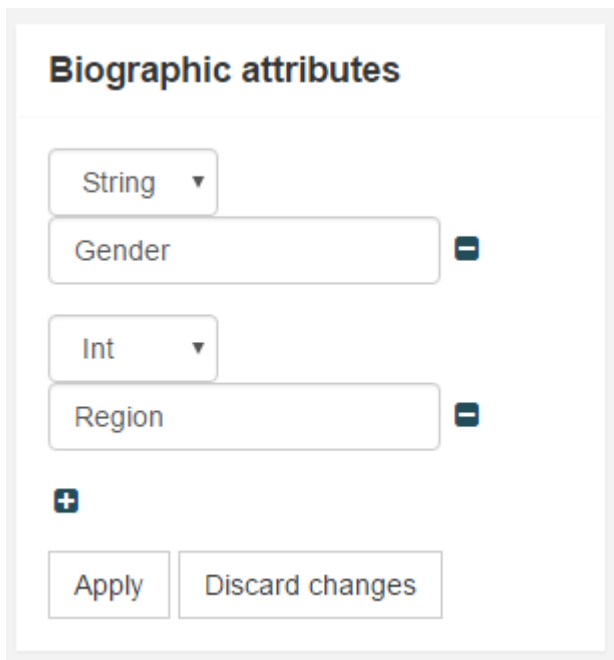
## Accelerated faces template size

Faces accelerator can operate in only one mode at a time:

- Choose large template size when using large or medium templates created with Neurotechnology SDK 11.0 or later (MegaMatcher/VeriLook). This option is recommended for new systems in order to achieve maximum recognition accuracy.
- Choose small when all or part of face templates were created with Neurotechnology SDKs 9.0 or 10.0. Such templates are NOT interoperable with NeurotecSDK 12.0 and later generated templates.

If upgrading from previous version, it is recommended to reextract face templates from original images

and choose the large template size. If reextraction is not possible and templates were created with SDKs 9.0 or 10.0, choose small option. Face templates created with Neurotec SDKs 6.0 or earlier version are not supported and will not be matched.



*Illustration 8: Biographic attributes configuration*

## Supervised Neurotechnology parameters

These parameters are for internal use. Do not change unless Neurotechnology support department has provided values optimized for specific case.

## Configure biographic attributes

Biographic attributes is any non-biometric data associated with a subject, such as gender, region, etc.

It can be used to filter subjects by non-biometric criteria in identification query. For example, identify only subjects residing in a specific region.

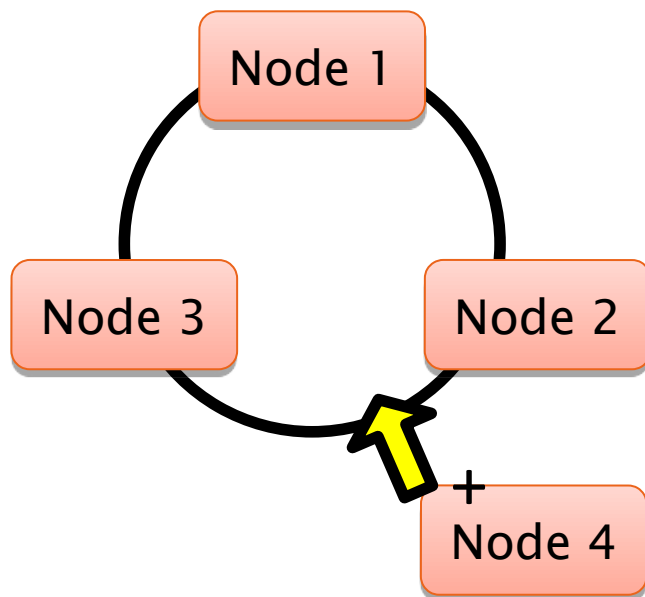
Biographic data is integrated with database support so it requires no additional effort to use except for specifying biographic data schema. MegaMatcher Accelerator biographic attributes scheme is specified using Configuration -> Biographic attributes option in Web Configuration.

User can specify up to 15 attributes as required. Illustration 8: Biographic attributes configuration above specifies biographic scheme with 2 attributes:

1. Gender (data type – String)
2. Region (data type – Int).

These attributes are saved by pressing Apply button.

## Adding a new node



*Illustration 9: Adding a new node to cluster scenario*

New node can be added to a running cluster without a need for downtime and even without interrupting existing work. Typically new nodes are added when system capacity or performance requirements increase. The following steps need to be performed:

1. The new node first has to be installed as described in chapter Replication. Do not start the node as yet.
2. In every already present node in the cluster run the add node command and enter the IP address of the new node using MMA Installation Wizard:  

```
# ./neurotec-biometric-node-ctl.sh add-node
```

3. Start the newly installed node as described in Start node.

The node will automatically download the necessary parts of the database from other nodes and take part of the workload when it has data loaded. No additional actions are required from administrator, except checking in Web Control Panel node loading status and ensuring that no unexpected failures have occurred.

## Backup and restore

There are two ways how backups can be created in MMA Cluster product: using standard database backup tools with helper shell scripts or using proprietary backup tool which supports some advanced scenarios.

### Database backup tools

MMA Cluster stores gallery of templates in one of the SQL databases. By default MariaDB is used for storage. Therefore standard tools for the SQL database can be used to create and restore backups. To make the task easier, helper shell scripts are provided which make it easy to perform common scenarios such as creating a full or incremental backup, and restore backup data on the system. The shell scripts and related `readme.txt` file which contains instructions how to perform any scenario step by step are provided in *backup-tools* folder in MMA installation.

Advanced system administrators may want to integrate backups directly into their workflow without using the provided helper shell scripts. In such case scripts can be used as an example to start with for custom integration.

Downside of using database backup method, is that in order to restore the data, it is required to stop the system and replace the existing database data with the copy from backup.

### Proprietary backup tool

Proprietary backup tool which is optimized for MMA Cluster data structure and can be used to perform advanced backup scenarios. Unlike database backup tool, restore procedure can be performed on running system. MMA only applies changes which are not already present in the gallery. If there is an interruption in restore process, it is possible to restart the process and it will continue from where it last finished. It is also possible to merge multiple cluster databases.

Proprietary backup tool can be accessed through webconfig user interface or web API.

# Support

Neurotechnology provides support for MegaMatcher Accelerator including:

- choosing optimal editions to meet project requirements,
- choosing optimal configuration to meet failover and high availability requirements,
- installing MMA on customers hardware,
- help with developing applications which send requests to MMA,
- operations and maintenance questions.

Contact support via email [support@neurotechnology.com](mailto:support@neurotechnology.com).